

solution 2 (resubmission)

Apoorv Shah

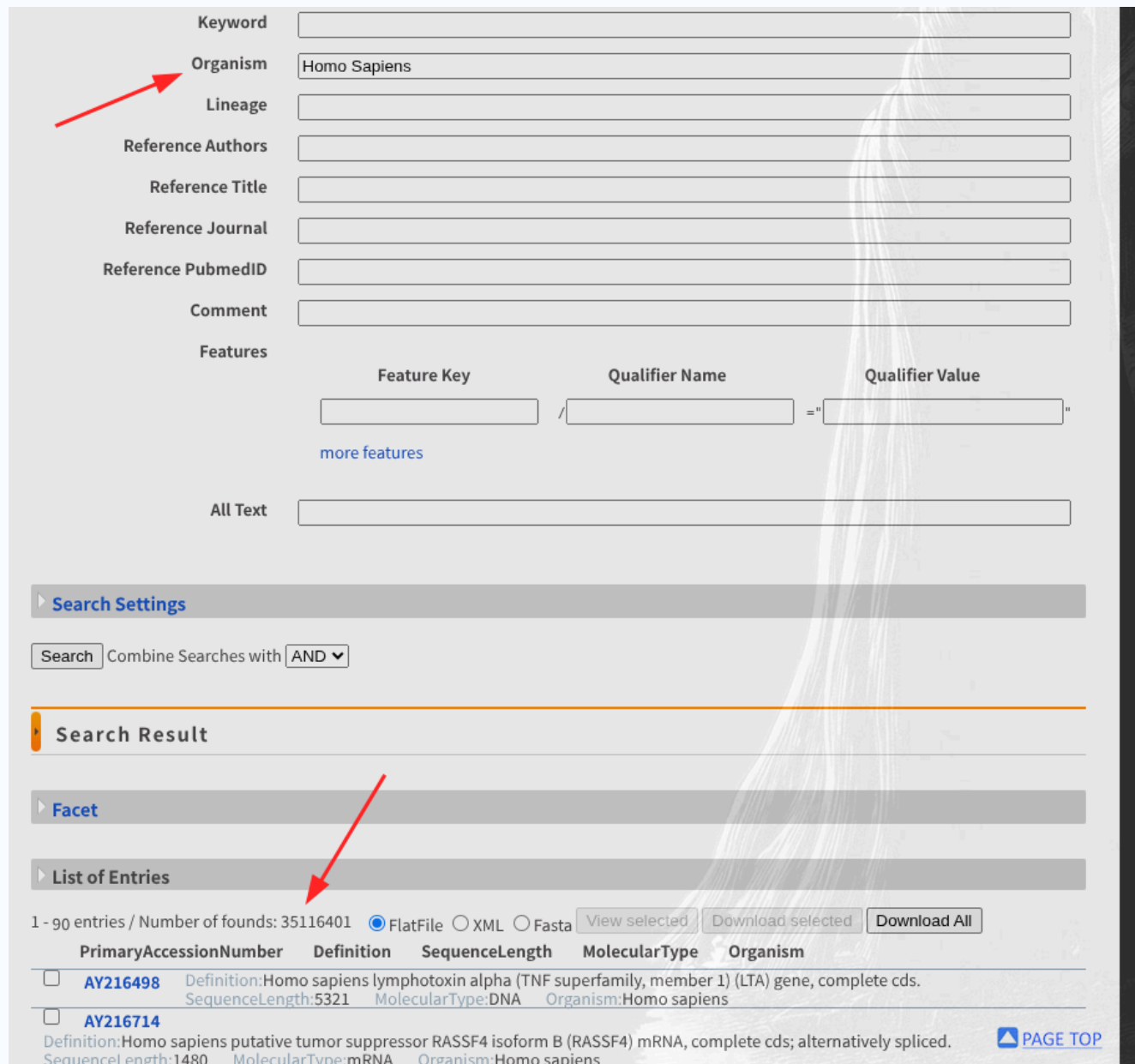
BS24B006

(sorry for sharing assignment, and extremely grateful for getting to resubmit)

Q1

DDBJ

this link opens the DDJB advanced search for *Homo sapiens*



Keyword

Organism

Lineage

Reference Authors

Reference Title

Reference Journal

Reference PubmedID

Comment

Features

Feature Key	Qualifier Name	Qualifier Value

[more features](#)

All Text

► Search Settings

Combine Searches with

► Search Result

► Facet

► List of Entries

1 - 90 entries / Number of founds: 35116401 ☒ FlatFile ☐ XML ☐ Fasta

PrimaryAccessionNumber	Definition	SequenceLength	MolecularType	Organism
☐ AY216498	Definition:Homo sapiens lymphotoxin alpha (TNF superfamily, member 1) (LTA) gene, complete cds.	SequenceLength:5321	MolecularType:DNA	Organism:Homo sapiens
☐ AY216714	Definition:Homo sapiens putative tumor suppressor RASSF4 isoform B (RASSF4) mRNA, complete cds; alternatively spliced.	SequenceLength:1480	MolecularType:mRNA	Organism:Homo sapiens

[PAGE TOP](#)

so DDBJ = 35,116,401

GenBank

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

Nucleotide Nucleotide Homo Sapiens[Organism] Search

Create alert Advanced Help

Species Summary 20 per page Sort by Default order

Animals (29,784,649)
Plants (1)
Fungi (1)
Viruses (4,548)
Customize ...

Molecule types

genomic
DNA/RNA (16,937,41)
mRNA (9,950,426)
rRNA (788)
Customize ...

Source databases

INSDC (GenBank) (29,256,885)
RefSeq (424,468)
Customize ...

Sequence Type

Items: 1 to 20 of 29841998

<< First < Prev Page 1 of 1492100 Next > Last >>

1. [Homo sapiens N-sulfoglucosamine sulfohydrolase \(SGSH\), RefSeqGene on chromosome 17](#)
26,661 bp linear DNA
Accession: NG_008229.2 GI: 2310189229
[Protein](#) [PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)

2. [Homo sapiens FASTK mitochondrial RNA regulator 3 \(FASTKD3\), transcript variant 3, non-coding RNA](#)
1,001 bp linear transcribed-RNA
Accession: NR_073608.2 GI: 1890392847
[PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)

Send to: Filters: [Manage Filters](#)

Results by taxon

Top Organisms [\[Tree\]](#)
Homo sapiens (29781320)
synthetic construct (57325)
Mus musculus (3176)
Human papillomavirus 16 (78)
Trypanosoma cruzi (21)
All other taxa (78)
[More...](#)

Find related data

Database: Select

Find items

Search details

in the nucleotide database in genbank it shows 29,841,998

EMBL

this link

ebi.ac.uk/ena/browser/advanced-search

Relaunch to update

Data Type Query Inclusion/Exclusion Fields Data Filters Results

Download ENA records: FASTA Download report: JSON

TEXT TSV

Accession	Description
A00383	H.sapiens NANB-hepatitis associated DNA
A06851	H.sapiens gene for relaxin, exon 1
A08002	Nucleotide sequence 7 from patent number EP0370719
A08905	H.sapiens (haplotype 2B, allele MS32, isolate English, serial number 31) minisatellite sequence
A11954	H.sapiens mRNA fragment for calcitonin
A14571	(2'-5') oligo A synthetase cDNA for 1.6kb RNA (E16,clone 9-21)
A25971	H.sapiens mRNA for T-cell receptor IGR b 20 Vbeta 9
A26249	Human shIL5R-alpha gene
A26437	Human granzyme B
A28104	Human GABAA receptor alpha-5 subunit

Items per page: 10 1 - 10 of 10000 (29244316 total)

New Search Back Copy Curl Request Copy Http Request

so 29,244,316

finally:

- ◆ DDBJ = 35,116,401
- ◆ GenBank = 29,841,998
- ◆ EMBL = 29,244,316

[New comment](#)

even though all three sync under INSDC every single day, the counts don't match up 1:1 mainly because of how each node

indexes raw ESTs/bulk reads, updates their search engines, and filters patent or patent-related sequences differently

Q.2

link

this is the sequence taken from genbank:

Plaintext

```
ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAAGGTG
TGAGTTGGCCAGAACTC
TGAAAAGATTGGAATGGATGGCTACAGGGGAATCAGCCTAGCAAACCTGGATG
TGTTTGGCCAAATGGGA
GAGTGGTTACAACACACGAGCTACAACTACAATGCTGGAGACAGAAGCACTG
ATTATGGGATATTTTCAG
ATCAATAGCCGCTACTGGTGTAATGATGGCAAAACCCCAGGAGCAGTTAATGC
CTGTCATTTATCCTGCA
GTGCTTTGCTGCAAGATAACATCGCTGATGCTGTAGCTTGTGCAAAGAGGGTT
GTCCGTGATCCACAAGG
CATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAACAGAGGTGTCCGTC
AGTATGTTCAAGGTTGT
GGAGTGTAATGA
```

using emboss we can figure out the GC content

GEECEE
Calculate fractional GC content of nucleic acid sequences

input section
Enter the 'seqall' input as:
☐ file/emboss-query or ☒ paste or ☐ list of files

Cut and Paste
 >AY330867
 ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGCTTTGAAAGGTGTGAGT

output section
 outfile file name

Execution mode:

AND ☐ OR ☐

File Colour
 12 Plain

ay330867.gjeecee cmd

#Sequence	GC content
AY330867	0.47

so it is 47%

[✎ New comment](#)

the total length of the sequence is 432 bp, and with G+C adding up to 203 bases, $(203 / 432)$ gives right around 0.4699, so 47% is correct

Q.3

- ◆ <https://www.ncbi.nlm.nih.gov/nuccore/33088589>
- ◆ https://getentry.ddbj.nig.ac.jp/getentry/na/AY330867/?format=flatfile&filetype=html&trace=true&show_suppressed=false&limit=10
- ◆ <https://www.ebi.ac.uk/ena/browser/view/AY330867?show=orcid-links>

all databases give the sequence, translation and other useful info

all three are part of the INSDC and sync daily, so the core biological sequences and accession numbers are identical across all of them

(taken from ddbj)

```

LOCUS      AY330867                      432 bp    mRNA    linear    SYN 29-JUL-2004
DEFINITION Synthetic construct human lysozyme mRNA, complete cds.
ACCESSION  AY330867
VERSION    AY330867.1
KEYWORDS   .
SOURCE     synthetic construct
  ORGANISM synthetic construct
            other sequences; artificial sequences.
REFERENCE  1 (bases 1 to 432)
AUTHORS    Xie,X.D., Chen,Z.H. and Zhou,C.S.
TITLE      Direct Submission
JOURNAL     Submitted (26-JUN-2003) Postdoctoral Scientific Research Station of
            Gansu Yasheng Industrial (Group) Co., Ltd., Zhangye Road, Lanzhou,
            Gansu 730030, P.R.China
FEATURES   Location/Qualifiers
     source          1..432
                     /organism="synthetic construct"
                     /mol\_type="mRNA"
                     /db\_xref="taxon:32630"
     CDS            1..429
                     /codon\_start=1
                     /transl\_table=11
                     /product="human lysozyme"
                     /protein\_id="AAP93336.1"
                     /translation="MRKKRRQRRRMKKVFERCELARTLKRLGMDGYRGISLANWMCLA
            KWESGYNTRATNYNAGDRSTDYGIQINSRYWCNDGKTPGAVNACHLSCSALLQDNIA
            DAVACAKRVVRDPQGIRAWVAWRNRCQNRGVRQYVQCGV"
BASE COUNT 131 a              72 c              129 g              100 t
ORIGIN
      1 atgaggaaga agcggagaca gcgacgaaga atgaagaagg tctttgaaag gtgtgagttg
     61 gccagaactc tgaaaagatt gggaatggat ggctacaggg gaatcagcct agcaaactgg
    121 atgtgtttgg ccaaattgga gagtgggttac aacacacgag ctacaaacta caatgctgga
    181 gacagaagca ctgattatgg gatatttcag atcaatagcc gctactgggtg taatgatggc
    241 aaaacccagg gagcagttaa tgcctgtcat ttatcctgca gtgctttgct gcaagataac
    301 atcgctgatg ctgtagcttg tgcaaaagagg gtgtccgtg atccacaagg cattagagca
    361 tgggtggcat ggagaaatcg ttgtcaaaac agaggtgtcc gtcagtatgt tcaaggttgt
    421 ggagtgaat ga
//

```

but they have some differences like:

- ◆ genbank and ddbj use full words like locus and origin for headers but embl uses two letter codes like id and sq
- ◆ ddbj clearly shows the exact base count right before the sequence but genbank skips it completely
- ◆ embl web view gives extra stuff like orcid data claims and cross references to other databases
- ◆ genbank and ddbj put the sequence numbers on the left side but embl puts them on the far right
- ◆ genbank it is tied to NCBI, meaning it natively cross-links to PubMed for literature, BLAST for alignments, and RefSeq for

curated genomes

- ◆ embl it is built into the european framework, directly linking nucleotide sequences to Ensembl (genome browsers) and UniProt (protein databases)
- ◆ ddbj it serves the asian research node and links out to specific functional databases like KEGG for metabolic pathway analysis

the raw sequence data is exactly the same; the real difference is just the web interface and the downstream analytical tools each database connects us to

 **New comment**

basically the actual FASTA/nucleotide string is identical line-for-line across all three, it's just the flatfile parser syntax and which bioinformatics ecosystem you get plugged into that changes between them

Q.4

papers link

The screenshot shows a Google Scholar search results page. The search query is "discrimination of beta barrel membrane proteins". The results are sorted by year. The first result is "A Hidden Markov Model method, capable of predicting and discriminating β -barrel outer membrane proteins" by PG Bagos, TD Liakopoulos, IC Spyropoulos, et al., published in BMC in 2004. The second result is "TMBETADISC-RBF: discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles" by YY Ou, MM Gromiha, SA Chen, M Suwa, published in Computational biology and chemistry in 2008. The third result is "Predicting transmembrane beta-barrels in proteomes" by HR Bigelow, DS Petrey, J Liu, D Przybylski, published in Nucleic acids research in 2004. The fourth result is "PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins" by KD Tsirigos, A Elofsson, PG Bagos, published in Bioinformatics in 2016. The fifth result is "A sequence-profile-based HMM for predicting and discriminating beta barrel membrane proteins" by PL Martelli, P Fariselli, A Krogh, R Casadio, published in ISMB in 2002. The sixth result is "Functional discrimination of membrane proteins using machine learning techniques" by MM Gromiha, Y Yabuki, published in BMC bioinformatics in 2008. The seventh result is partially visible: "TMB-Hunt: an amino acid composition-based method to screen proteomes for...".

some examples:

- ◆ Liu, Q., Zhu, Y., Wang, B., & Li, Y. (2003). Identification of β -barrel membrane proteins based on amino acid composition properties and predicted secondary structure. *Computational Biology and Chemistry*, 27, 355–361. [https://doi.org/10.1016/s1476-9271\(02\)00085-3](https://doi.org/10.1016/s1476-9271(02)00085-3) Cited by: 52
- ◆ Ou, Y.-Y., Gromiha, M. M., Chen, S.-A., & Suwa, M. (2008). TMBETADISC-RBF: Discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles. *Computational Biology and Chemistry*, 32, 227–231.

<https://doi.org/10.1016/j.compbiolchem.2008.03.002> Cited by: 106

Pubmed link as well

The screenshot shows the PubMed interface. At the top, there's a header for the NIH National Library of Medicine. The search bar contains the query "discrimination of beta barrel membrane proteins". Below the search bar, there are options for "Advanced", "Create alert", and "Create RSS". The search results section shows "54 results" and "2 articles found by citation matching". The first article listed is "TMBETADISC-RBF: Discrimination of beta-barrel membrane proteins using RBF networks and PSSM profiles." by Ou YY, et al. (2008). The second article is "Current developments on beta-barrel membrane proteins: sequence and structure analysis, discrimination and prediction." by Gromiha MM, et al. (2007). There is a warning message: "Quoted phrase not found in phrase index: 'discrimination of beta barrel membrane proteins'". On the left side, there are filters for "MY CUSTOM FILTERS", "RESULTS BY YEAR" (a bar chart from 1992 to 2026), "PUBLICATION DATE" (1 year, 5 years, 10 years, Custom Range), and "TEXT AVAILABILITY" (Abstract, Free full text).

- ◆ Tsirigos, K. D., Elofsson, A., & Bagos, P. G. (2016). PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins. *Bioinformatics*, 32, i665–i671.

<https://doi.org/10.1093/bioinformatics/btw444> Cited by: 103

[New comment](#)

predicting beta-barrels is way harder than alpha-helical proteins because the hydrophobic pattern alternates every single residue (facing lipid vs lumen) rather than staying in continuous hydrophobic stretches

Q.5

Kihara D Google Scholar link

one paper published in nature journal example

paper link

they report the results from the first large-scale community-based critical assessment of protein function annotation (CAFA) experiment.

on pubmed

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NIH National Library of Medicine
National Center for Biotechnology Information

Log in

PubMed Advanced Search Builder

PubMed
User Guide

Add terms to the query box

Author Kihara D. X

ADD

Show Index

Query box

Enter / edit your search query here

Search

An official website of the United States government [Here's how you know](#) ✓

NIH National Library of Medicine
National Center for Biotechnology Information

PubMed

Kihara D.[Author] × **Search**

[Advanced](#) [Create alert](#) [Create RSS](#) [User Guide](#)

[Filters](#) [Timeline](#) Sort by: [Best match](#) [Display options](#)

[Save](#) [Email](#) [Send to](#)

275 results « < Page 1 of 28 > »

i It looks like you are searching for an author. Results are currently sorted by Best Match. To see the newest results first, [change the sort order to Most Recent](#). ×

☐ [Effect of phosphorylation barcodes on arrestin binding to a chemokine receptor.](#)
1 Chen Q, Schafer CT, Mukherjee S, Wang K, Gustavsson M, Fuller JR, Tepper K, Lamme TD, Aydin Y,
Cite Agrawal P, Terashi G, Yao XQ, **Kihara D**, Kossiakoff AA, Handel TM, Tesmer JG.
Nature. 2025 Jul;643(8070):280-287. doi: 10.1038/s41586-025-09024-9. Epub 2025 May 21.
PMID: 40399676 [Free PMC article](#).

☐ [Computational protein function predictions.](#)
2 **Kihara D.**
Cite Methods. 2016 Jan 15;93:1-2. doi: 10.1016/j.ymeth.2016.01.001.
PMID: 26778120 No abstract available.

☐ [Computational prediction of disordered binding regions.](#)
3 **Kihara D.**

[New comment](#)

CAFA is basically the CASP equivalent for automated protein function prediction, testing algorithms blindly across Gene Ontology categories (molecular function, biological process, cellular component)

Q.6

paper link

"Mitochondrial beta-barrel proteins, an exclusive club?"

there are lots of related articles

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NIH National Library of Medicine
National Center for Biotechnology Information

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PubMed

Cell 2008 Dec 26;135(7):1158-9? × **Search**

[Advanced](#) [Create alert](#) [Create RSS](#) [User Guide](#)

Found 1 result for an alternative search.
Your search for *Cell 2008 Dec 26;135(7):1158-9?* retrie...

[Save](#) [Email](#) [Send to](#) [Display options](#)

[Comment](#) > [Cell. 2008 Dec 26;135\(7\):1158-9; author reply 1159-60.](#)
doi: 10.1016/j.cell.2008.12.017.

Mitochondrial beta-barrel proteins, an exclusive club?

[Kenichiro Imai, M Michael Gromiha, Paul Horton](#)

PMID: 19109882 DOI: [10.1016/j.cell.2008.12.017](#)

Free article

[Full text links](#) [Cite](#)

No abstract available

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Comment on

Similar articles

[Transport proteins \(carriers\) of mitochondria.](#)

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

[Systematic analysis of the twin cx\(9\)c protein family.](#)

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.

PMID: 19703468

[Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.](#)

Javadov S, Karmazyn M, Escobales N.

J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.

PMID: 19509316

[Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.](#)

de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.

Toxicol Appl Pharmacol. 2009 Sep 15;239(3):241-50. doi: 10.1016/j.taap.2009.05.028. Epub 2009 Jun 11.

PMID: 19523970

[A CaPful of mechanisms regulating the mitochondrial permeability transition.](#)

Di Lisa F, Bernardi P.

J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.

PMID: 19303419 Review.

[See all similar articles](#)

Total 27 similar articles are listed here
on pubmed as well

Comment on
Full text links

Cell. 2008 Mar 21;132(6):1011-24. doi: 10.1016/j.cell.2008.01.028.
PMID: 18358813

Similar articles

Transport proteins (carriers) of mitochondria.
Wohlrab H.
IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.
PMID: 18816452 Review.

Systematic analysis of the twin cx(9)c protein family.
Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.
J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.
PMID: 19703468

Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.
Javadov S, Karmazyn M, Escobales N.
J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.
PMID: 19509316

Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.
de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.
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PMID: 19523970

A CaPful of mechanisms regulating the mitochondrial permeability transition.
Di Lisa F, Bernardi P.
J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.
PMID: 19303419 Review.

[See all similar articles](#)

New comment

PubMed groups these using their text-neighboring algorithm (looking at shared MeSH terms and title/abstract keywords), which is why most of the 27 results focus on the SAM complex and outer mitochondrial membrane insertion

Q.7

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NIH National Library of Medicine
National Center for Biotechnology Information

PubMed

Nature[Journal] AND 2026[Date - Publication]

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Sort by:

MY CUSTOM FILTERS
[Edit custom filters](#)

2,937 results

Page 1 of 294

RESULTS BY YEAR

PUBLICATION DATE

☒ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

☒ Filters applied: in the last 1 year. [Clear all](#)

- ☐ **Ketogenic diet mediates intestinal tumorigenesis through lipids not ketones.**
 Shay JES, Chi F, Tzouanas CN, Han S, Zhang X, Ten Hoeve J, Williams KJ, Neptun S, Sever T, Fuentes I, Bhatia SN, Calibasi-Kocal G, Vander Heiden MG, Shalek AK, Yilmaz ÖH.
 Cite **Nature.** 2026 Aug;656(8127):474-483. doi: 10.1038/s41586-026-10779-y. Epub 2026 Jul 15. PMID: 42457955
- ☐ **GPC3-specific dnTGFβRII-armoured CAR T cells for hepatocellular carcinoma.**
 Zhang Q, Fu Q, Shen Y, Cao W, Jin G, Zhang Y, Wu J, Chen C, Wang H, Xu X, Sun K, Xue X, Bondanza A, Cao D, Chu N, Durham N, Cobbold M, Farsaci B, Giardino Torchia ML, Hao Y, Hong Y, Huang J, Humphries M, Jian Q, Moody G, Stone J, Sun L, Tu E, Wang F, Wang F, Yang Y, Yao Y, Zou A, Bai X, Liang T.
 Cite **Nature.** 2026 Jul 15. doi: 10.1038/s41586-026-10786-z. Online ahead of print. PMID: 42457964
- ☐ **Diet-microbiome synergy underlies obesity-associated immunotherapy efficacy.**
 Desharnais L, Swaby A, Messaoudene M, Doré S, Yu MW, Fiset B, Breton V, Ponce M, Hu Y, Wilson L, Sorin M, Wang Y, Dewar K, Pollak M, Elrief A, Routy B, Walsh LA, Quail DF.
 Cite **Nature.** 2026 Jul 8. doi: 10.1038/s41586-026-10750-x. Online ahead of print. PMID: 42420462
- ☐ **Mitochondrial metabolism and epigenetic crosstalk drive SASP.**
 Martini H, Birch J, Marques EDM, Victorelli S, Lannado AB, Pirius N, Franco AC, Lee G, Han Y, Rowsey J, II

Pubmed shows 2937 articles

SCOPUS shows 22,924 articles

New comment

the huge jump in Scopus makes sense since PubMed strictly indexes life sciences and biomedical journals, while Scopus indexes cross-disciplinary engineering, materials science, and physics literature as well

Q.8

Burkhard Rost
 Professor of Computer Science, [Technical University Munich](#)
 Verified email at tum.de - [Homepage](#)
[machine learning](#) [artificial intelligence](#) [bioinformatics](#) [computer science](#) [protein prediction](#)

TITLE	CITED BY	YEAR
The transcriptional landscape of the mammalian genome P Carninci, T Kasukawa, S Katayama, J Gough, MC Frith, N Maeda, ... science 309 (5740), 1559-1563	4251	2005
Prediction of protein secondary structure at better than 70% accuracy B Rost, C Sander Journal of molecular biology 232 (2), 584-599	3970	1993
ProtTrans: toward understanding the language of life through self-supervised learning	3809 *	2021

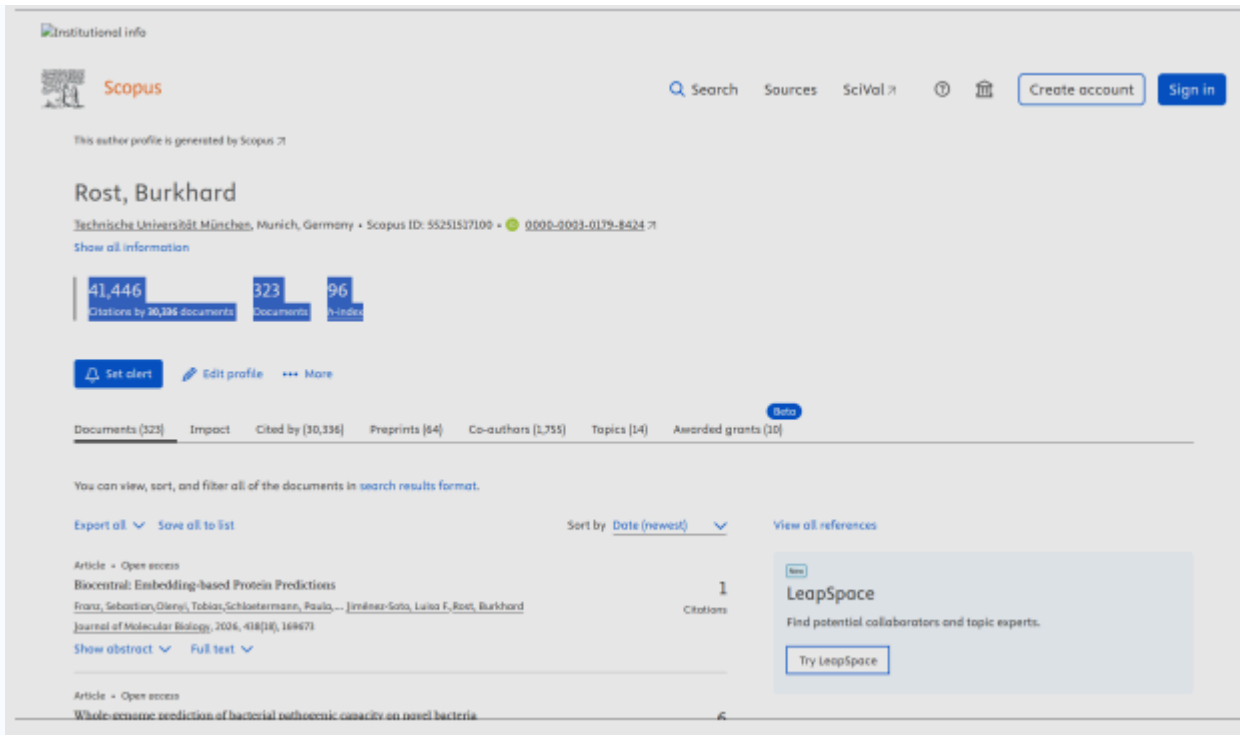
Cited by [VIEW ALL](#)

	All	Since 2021
Citations	65146	19704
h-index	116	60
i10-index	292	182

Bar chart showing citations over time (2015-2021):

Year	Citations
2015	~2000
2016	~2500
2017	~2800
2018	~3000
2019	~3200
2020	~3500
2021	~4400

on google scholar



Institutional info

Scopus

This author profile is generated by Scopus.™

Rost, Burkhard

Technische Universität München, Munich, Germany • Scopus ID: 5525157100 • 0000-0002-0179-8624

Show all information

41,446 Citations by 30,336 documents | 323 Documents | 96 h-index

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Documents (323) | Impact | Cited by (30,336) | Preprints (84) | Co-authors (3,735) | Topics (14) | Awarded grants (10)

You can view, sort, and filter all of the documents in search results format.

Export all | Save all to list | Sort by Date (newest) | View all references

Article • Open access
 Biocentral: Embedding-based Protein Predictions
 Franz, Sebastian; Olariu, Tobias; Schaefermann, Paula; ... Jiménez-Goto, Luisa F.; Rost, Burkhard
 Journal of Molecular Biology, 2026, 418(18), 169673
 Show abstract | Full text

Article • Open access
 Whole-genome prediction of bacterial pathogenic capacity on novel bacteria

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on pubmed

41,446 citations and 96 h-index

New comment

an h-index of 96 means at least 96 of the papers have been cited 96 or more times, which is a massive benchmark in computational structural biology

Q.9

link to brenda

Information on EC 1.7.2.3 - trimethylamine-N-oxide reductase

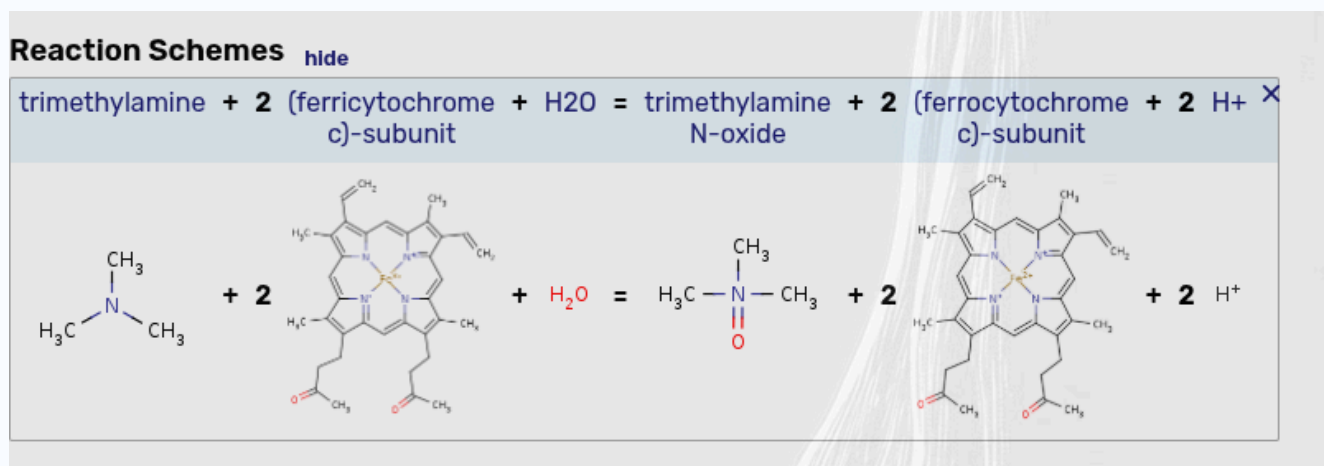
for references in articles please use BRENDA:EC1.7.2.3

EC Tree → 1 Oxidoreductases → 1.7 Acting on other nitrogenous compounds as donors → 1.7.2 With a cytochrome as acceptor → 1.7.2.3 trimethylamine-N-oxide reductase

this shows the enzyme class hierarchy

- ◆ Accepted Name: Trimethylamine-N-oxide reductase (cytochrome c), also commonly known as TMAO reductase.
- ◆ Enzyme Class Hierarchy:
 - ◆ EC 1 (Class): Oxidoreductases (enzymes that catalyze oxidation-reduction reactions).
 - ◆ EC 1.7 (Subclass): Acting on other nitrogenous compounds as donors.
 - ◆ EC 1.7.2 (Sub-subclass): With a cytochrome as an acceptor.
- ◆ Function / Catalytic Activity: This microbial enzyme catalyzes the chemical reduction of trimethylamine N-oxide (TMAO) into trimethylamine (TMA) as part of the electron transport chain.

reaction



[New comment](#)

bacteria like *E. coli* use this as an alternative terminal electron acceptor under anaerobic respiration when oxygen isn't around

Q.10

finding the EC number on brenda

6.3.5.4

asparagine synthase (glutamine-hydrolysing)

now searching it on ebi(thornton) gives us the enzyme

<https://www.ebi.ac.uk/thornton-srv/m-csa/entry/302/>

PDB

[1ct9Q](#) - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å)  

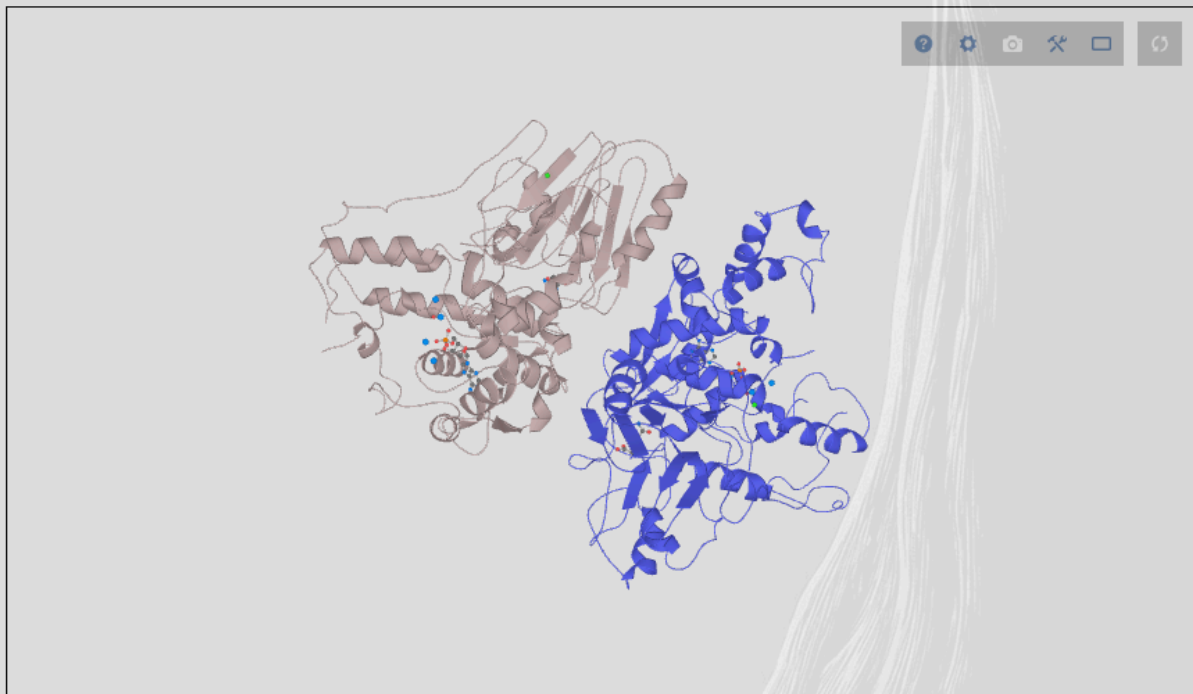


Catalytic CATH Domains

[3.60.20.10Q](#)  [3.40.50.620Q](#)  (see all for [1ct9Q](#))

Cofactors

Magnesium(2+) (1) [Metal MACiE](#)



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and below it shows the catalytic residues and their roles

Catalytic Residues Roles

UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

so answer = Cys 1, Cys 2, Gly76, Leu 51, Thr 322, and Arg 325.

New comment

the Cys residues are primarily doing the heavy lifting for coordinating the molybdenum/iron-sulfur active center, while the others stabilize the transition state and hydrogen-bond with the TMAO substrate

Q.11

Search for as complete name

Homo sapiens

Taxonomy ID: 9606 (for references in articles please use ncbitaxon:9606)

current name

Homo sapiens Linnaeus, 1758

Genbank common name: **human**

NCBI BLAST name: **primates**

Rank: **species**

Genetic code: [Translation table 1 \(Standard\)](#)

Mitochondrial genetic code: [Translation table 2 \(Vertebrate Mitochondrial\)](#)

Lineage(full)

[cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Deuterostomia](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Gnathostomata](#); [Teleostomi](#); [Euteleostomi](#); [Serraniformes](#)

Felis catus

Taxonomy ID: 9685 (for references in articles please use ncbitaxon:9685)

current name

Felis catus Linnaeus, 1758

includes: *Korat cats* L.

homotypic synonym:
Felis silvestris catus

Genbank common name: **domestic cat**

just listing the data below now:

Organism	Scientific Name	Taxonomy ID	Number of Chromosomes (2n)
Human	<i>Homo sapiens</i>	9606	46
Cat	<i>Felis catus</i>	9685	38
Dog	<i>Canis lupus familiaris</i>	9615	78
Domestic guinea pig	<i>Cavia porcellus</i>	10141	64
Thale cress	<i>Arabidopsis thaliana</i>	3702	10

[New comment](#)

Arabidopsis having only $2n = 10$ (5 chromosome pairs) is one of the main reasons it became the universal model organism for plant genetics, since the genome is super compact

Q.12

The Entrez Programming Utilities (E-utilities) are a suite of server-side programs that serve as the public API for the NCBI Entrez system. They allow users to programmatically search, link, and retrieve data from across all NCBI Entrez databases (including PubMed, Nucleotide, Protein, and Gene) without needing to use the web browser interface.

example syntax:

`esearch.fcgi?db=<database>&term=<query>`

example:

- ◆ [https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?](https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science)
[db=pubmed&term=science](https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science) 
 [journal]+AND+breast+cancer+AND+2008[pdat]

to fetch a fasta format:

[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?](https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text)
[db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text](https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text) 

syntax is:

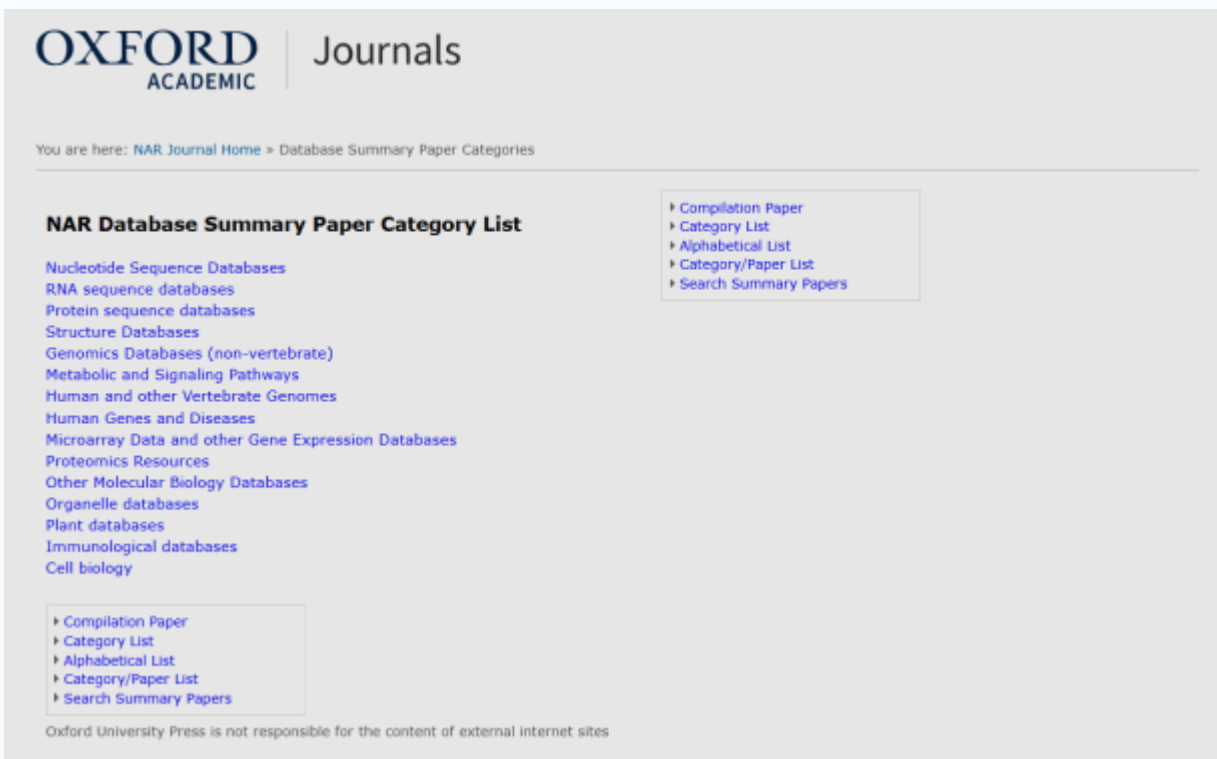
- ◆ `db=nuccore`: Specifies the database (e.g., `nuccore` for nucleotide, or `protein`).
- ◆ `id=AY330867`: The unique identifier or accession number of the record.
- ◆ `rettype=fasta`: Tells the server you want the specific FASTA layout.
- ◆ `retmode=text`: Tells the server to return plain text rather than XML.

New comment

E-utilities has a default rate limit of 3 requests/sec without an NCBI API key (or 10 requests/sec with a key), which is why programmatic pipelines usually cache UIDs using **esearch** first before batch downloading them with **efetch**

Q.13

a. Protein properties



The screenshot shows the Oxford Academic Journals website. The header includes the Oxford Academic logo and the word "Journals". Below the header, a breadcrumb trail reads "You are here: NAR Journal Home » Database Summary Paper Categories". The main content area is titled "NAR Database Summary Paper Category List". It contains a list of database categories: Nucleotide Sequence Databases, RNA sequence databases, Protein sequence databases, Structure Databases, Genomics Databases (non-vertebrate), Metabolic and Signaling Pathways, Human and other Vertebrate Genomes, Human Genes and Diseases, Microarray Data and other Gene Expression Databases, Proteomics Resources, Other Molecular Biology Databases, Organelle databases, Plant databases, Immunological databases, and Cell biology. To the right of this list, there is a box containing a list of options: Compilation Paper, Category List, Alphabetical List, Category/Paper List, and Search Summary Papers. At the bottom of the page, a small disclaimer states: "Oxford University Press is not responsible for the content of external internet sites".

General sequence databases

Protein properties

AAindex
 BindingDB
 Cybase
 dbPTM
 eProS
 HHMD
 IProLINK
 MALISAM
 MegaMotifbase
 MemMoRF
 Minimotif miner
 MobiDB
 MP:PD
 PFD - Protein Folding Database
 PIDD
 PINT
 PPD
 PPT-DB
 Proteome-pI 2.0
 ProTherm
 PRTAD
 REFOLD
 SwissSidechain
 TOPPR

1. UniProtKB (Universal Protein Resource): The central hub for the collection of functional information on proteins, with accurate, consistent, and rich annotation of protein properties.
2. Pfam (or InterPro): A large collection of protein families, represented by multiple sequence alignments and hidden Markov models (HMMs), used to determine protein domain properties.

b. Small molecules (Structure related)

Transcriptional regulator sites and transcription factors

Small molecules

BitterDB
 ChEBI - Chemical Entities of Biological Interest
 ChemBank
 ChemDB
 Crystallography Open Database
 CSD - Cambridge Structural Database
 EDULISS (currently unavailable)
 FragmentStore
 Hemolytik
 Het-PDB Navi
 HIC-Up
 MMsINC
 mVOC
 PDB-Ligand
 PubChem
 R.E.DD.B.
 SCRIPDB
 SuperDrug
 SuperScent
 SuperToxic

1. PubChem: A massive database maintained by NCBI that provides information on the chemical structures, physical properties, and biological activities of small molecules
2. ChEBI (Chemical Entities of Biological Interest): A freely available dictionary of molecular entities focused on strictly "small" chemical compounds and their structural properties.

c. Cancer gene databases

Cancer gene databases

ArrayMap
BCCTBbp
BreCAN-DB
Cancer RNA-Seq Nexus
Cancer3D
CancerGenes
CancerPPD
Candidate Cancer Gene Database
CanGEM
CanSAR
CaSNP
cBioPortal
CCDB
ccmGDB
CellLineNavigator
ChimerDB
CIVICdb
ClinVar
CMPD
Colorectal Cancer Atlas
COLT-Cancer
COSMIC
CTDatabase
dbDEMC
dbDEPC
DIVAS
DriverDBv2
FusionCancer
HLungDB
HPTAA
IARC TP53 Database
IGDB.NSCLC
intoGen
ITTACA
Inc2cancer
MethHC
MethyCancer
MoKCa
MutationAligner
Network of Cancer Genes
OncoDB.HCC
OpenTein
Pancreas Expression
PRECOG and related resources
rSNPBase 3.0

1. COSMIC (Catalogue of Somatic Mutations in Cancer): The world's largest and most comprehensive resource for exploring the impact of somatic mutations in human cancer.
2. cBioPortal for Cancer Genomics: An open-access resource for interactive exploration of multidimensional cancer genomics data sets.

 [New comment](#)

COSMIC focuses on curated somatic point mutations and gene fusions per sample, whereas cBioPortal is great for visualizing clinical patient cohorts with copy number alterations (CNA) and RNA expression side by side

Class Assignment

Question 1

Sequences:

- ◆ Sequence 1: ACAGACGTCCG (11 bases)
- ◆ Sequence 2: ACCGTCCG (8 bases)

Scoring Parameters:

- ◆ Match: +2
- ◆ Mismatch: -1
- ◆ Gap: -2

Recurrence Formulas:

- ◆ Diagonal: $F(i-1, j-1) + (+2 \text{ if match, } -1 \text{ if mismatch})$
- ◆ Up (Gap in Sequence 2): $F(i-1, j) - 2$
- ◆ Left (Gap in Sequence 1): $F(i, j-1) - 2$
- ◆ Cell Score: $F(i, j) = \max(\text{Diagonal}, \text{Up}, \text{Left})$

Generated by <https://babcock.net>

_	**_**	**A**	**C**	**C**	**G**	**T**	**C**	**C**	**G**
_	**0**	-2	-4	-6	-8	-10	-12	-14	-16
A	-2	**2**	0	-2	-4	-6	-8	-10	-12
C	-4	0	**4**	2	0	-2	-4	-6	-8
A	-6	-2	**2**	3	1	-1	-3	-5	-7
G	-8	-4	**0**	1	5	3	1	-1	-3
A	-10	-6	**2**	-1	3	4	2	0	-2
C	-12	-8	-4	**0**	1	2	6	4	2
G	-14	-10	-6	-2	**2**	0	4	5	6
T	-16	-12	-8	-4	0	**4**	2	3	4
C	-18	-14	-10	-6	-2	2	**6**	4	2
C	-20	-16	-12	-8	-4	0	4	**8**	6
G	-22	-18	-14	-10	-6	-2	2	6	**10**

Traceback:

Start from the bottom-right cell (10) and trace backward through the cells that yielded each optimal score until reaching the origin (0, 0).

optimal global alignment:

Plaintext

```

A C A G A C G T C C G
A C - - - C G T C C G

```

[New comment](#)

the 3 gaps in Sequence 2 (-6 penalty) are necessary because matching the entire tail **CGTCCG** yields +12, which heavily outweighs introducing a gap penalty across **AGA**

Question 2

The sequences are:

- ◆ Sequence 1: ACGTATCGCCGTATA (15 bases)
- ◆ Sequence 2: GATGCTCTCCGAAA (14 bases)

Scoring parameters:

- ◆ Match: +2
- ◆ Mismatch: -1
- ◆ Gap: -2

For the Smith-Waterman formula,

0 is included as a lower bound because this is a local alignment. If continuing the alignment produces a negative score, the score resets to 0 and a new alignment can begin at that position:

$$H(i, j) = \max(0, H(i-1, j-1) + \text{score}, H(i-1, j) - 2, H(i, j-1) - 2)$$

** *	** *	**G**	**A**	**T**	**G**	**C**	**T**	**C**	**T**	**C**	**C**	**G**	**A**	**A**	**A**
-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	0	2	0	0	0	0	0	0	0	0	0	2	2	2
C	0	0	0	1	0	2	0	2	0	2	2	0	0	1	1
G	0	2	0	0	3	1	1	0	1	0	1	4	2	0	0
T	0	0	1	2	1	2	3	1	2	0	0	2	3	1	0
A	0	0	2	0	1	0	1	2	0	1	0	0	4	5	3
T	0	0	0	4	2	0	2	0	4	2	0	0	2	3	4
C	0	0	0	2	3	4	2	4	2	6	4	2	0	1	2
G	0	2	0	0	4	2	3	2	3	4	5	6	4	2	0
C	0	0	1	0	2	6	4	5	3	5	6	4	5	3	1
C	0	0	0	0	0	4	5	6	4	5	7	5	3	4	2
G	0	2	0	0	2	2	3	4	5	3	5	9	7	5	3
T	0	0	1	2	0	1	4	2	6	4	3	7	8	6	4
A	0	0	2	0	1	0	2	3	4	5	3	5	9	10	8
T	0	0	0	4	2	0	2	1	5	3	4	3	7	8	9
A	0	0	2	2	3	1	0	1	3	4	2	3	5	9	10

For Needleman–Wunsch, the traceback starts at the bottom-right corner. For Smith–Waterman, traceback starts at the highest score anywhere in the table (10) and stops when a cell with a score of 0 is reached.

optimal local alignment:

Plaintext

TCGCCGTATA

TCTCCG-AAA

[New comment](#)

here the alignment naturally cuts off the noisy N-terminal overhangs (**ACGTA** in Seq 1 and **GATG** in Seq 2) since local alignment discards regions that drag the cumulative score down